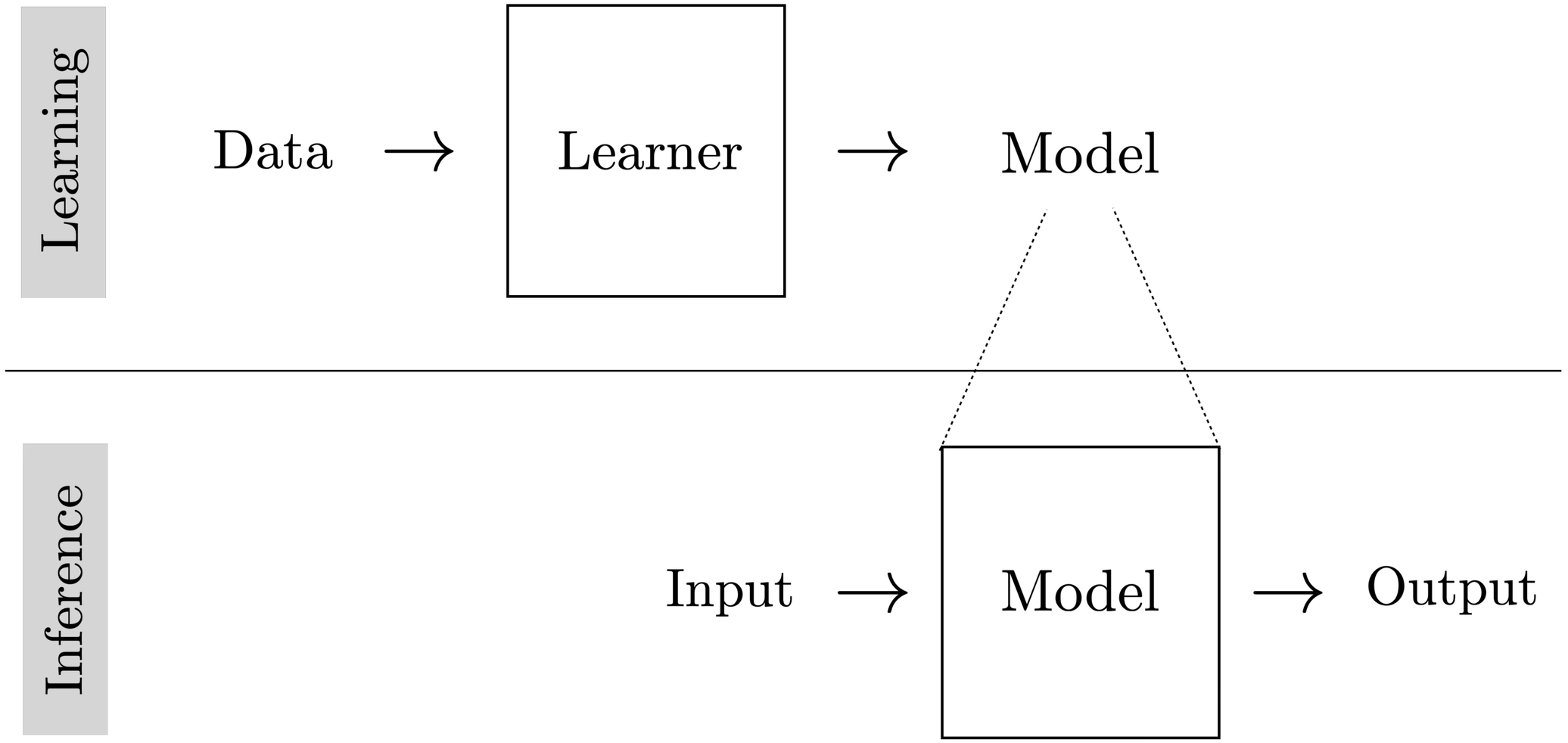
西南交通大学
Southwest Jiaotong University

SCAI011112 视觉计算

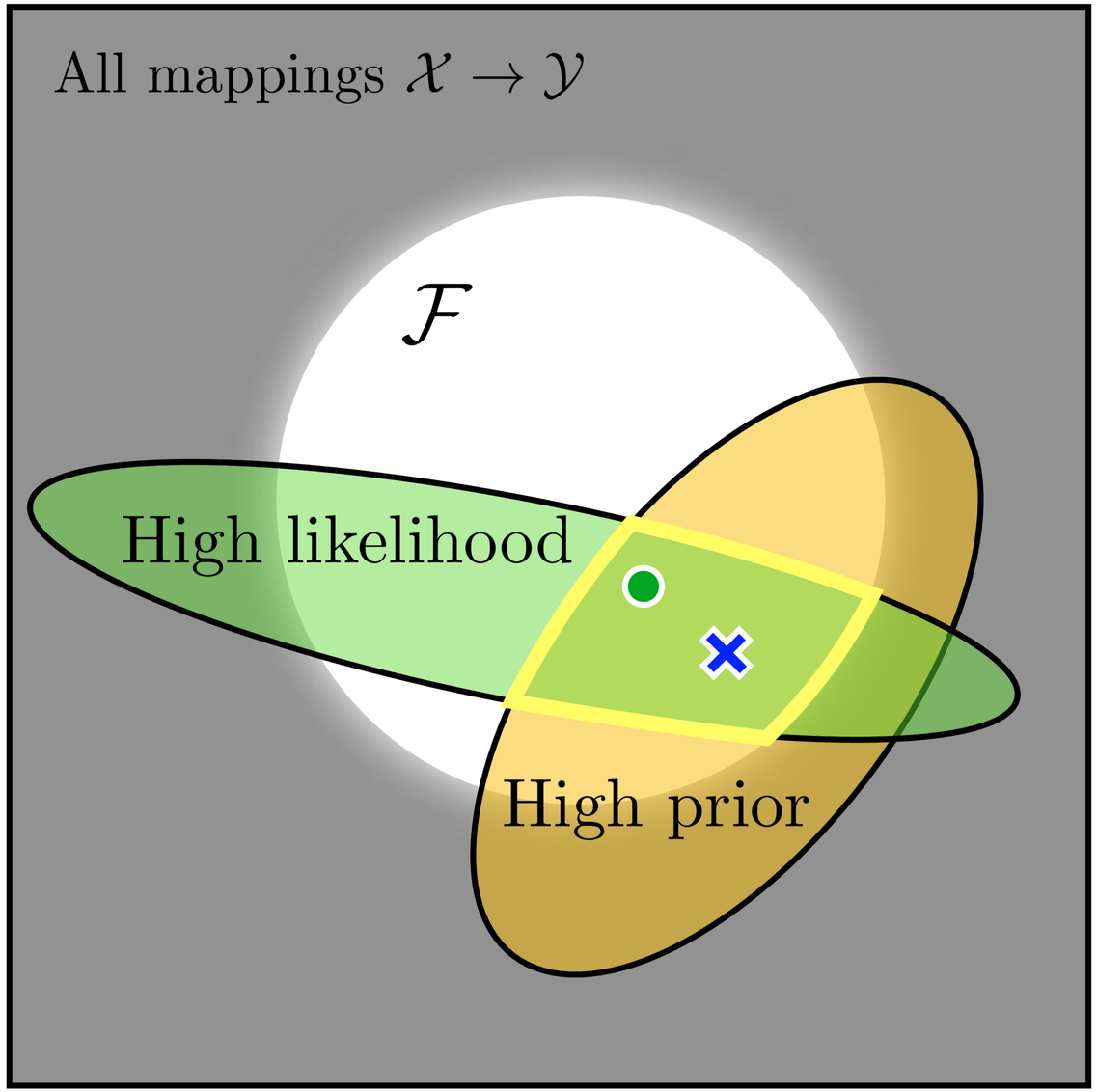
回顾：神经网络

计算机科学与技术专业
2025-2026(2)

机器学习



Data, Priors, and Hypotheses

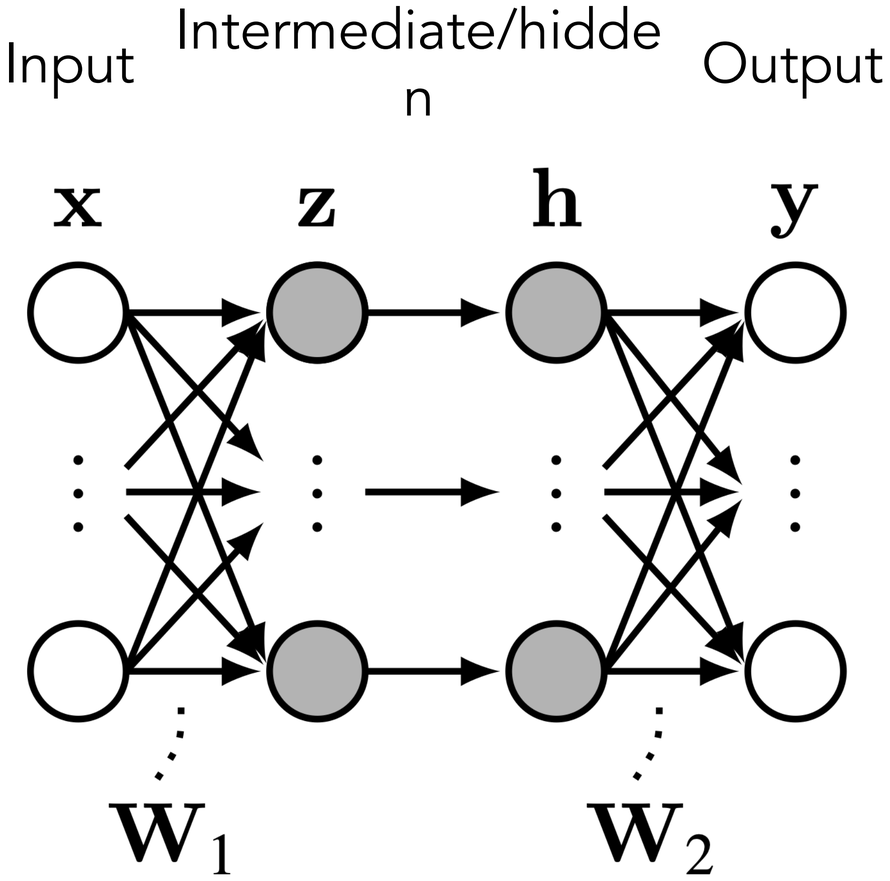


$\mathcal{F}$ – Hypothesis space

● True solution

✖ Learned solution

Multilayer Perceptron (MLP)

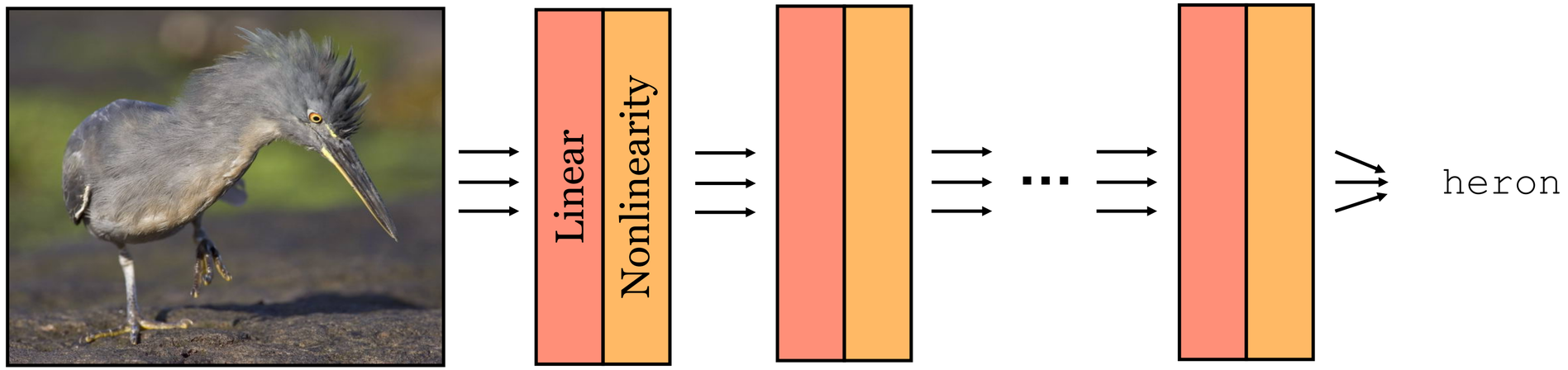


$$\mathbf{h} = g(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)$$

$$\mathbf{y} = g(\mathbf{W}_2\mathbf{h} + \mathbf{b}_2)$$

$$\theta = \{\mathbf{W}_1, \ldots, \mathbf{W}_L, \mathbf{b}_1, \ldots, \mathbf{b}_L\}$$

深度网络

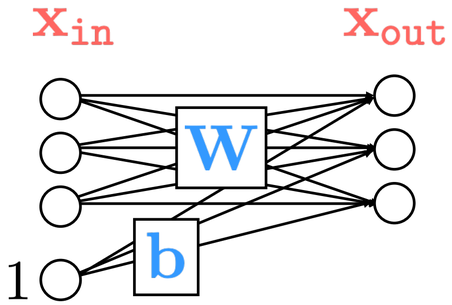


Wiring graph

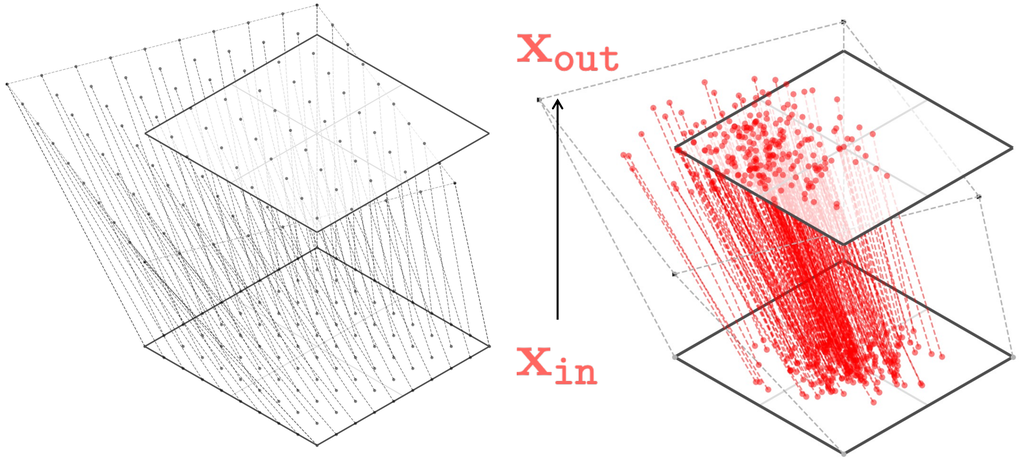
Equation

Mapping

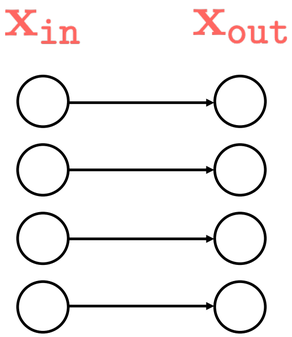
linear



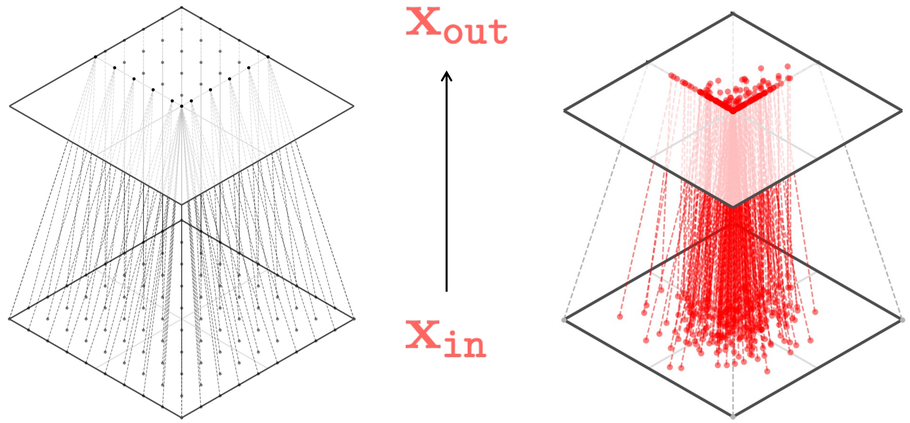
$$x_{out} = Wx_{in} + b$$



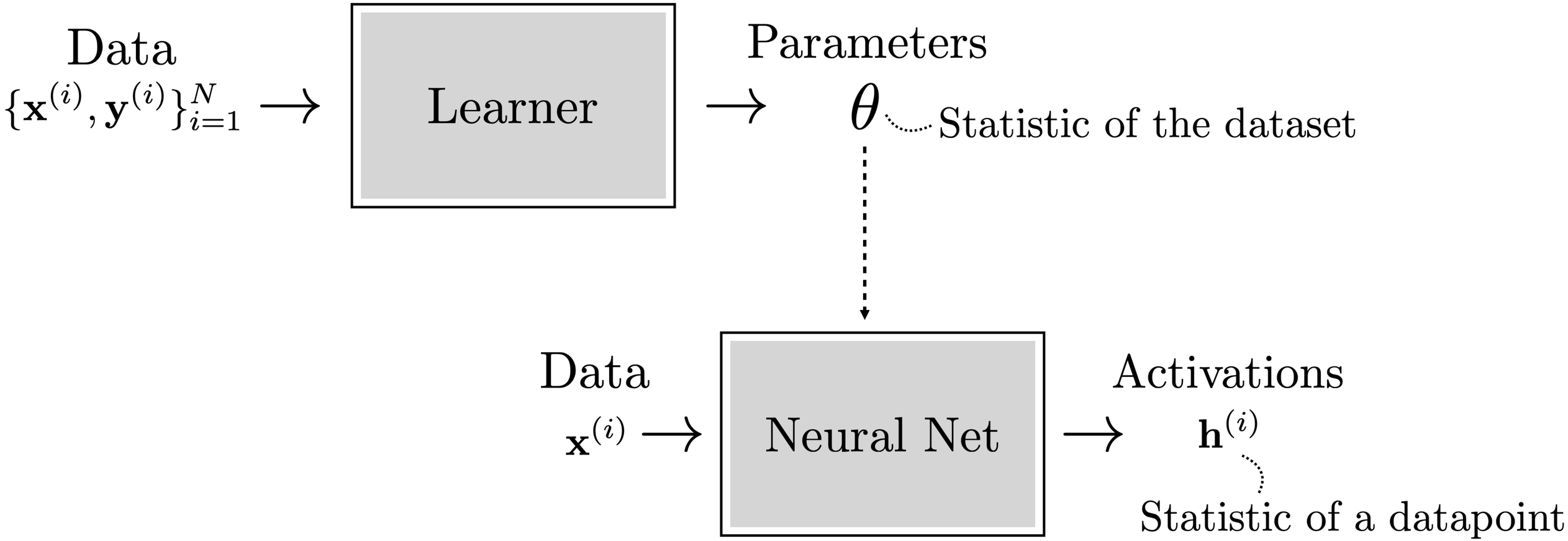
relu

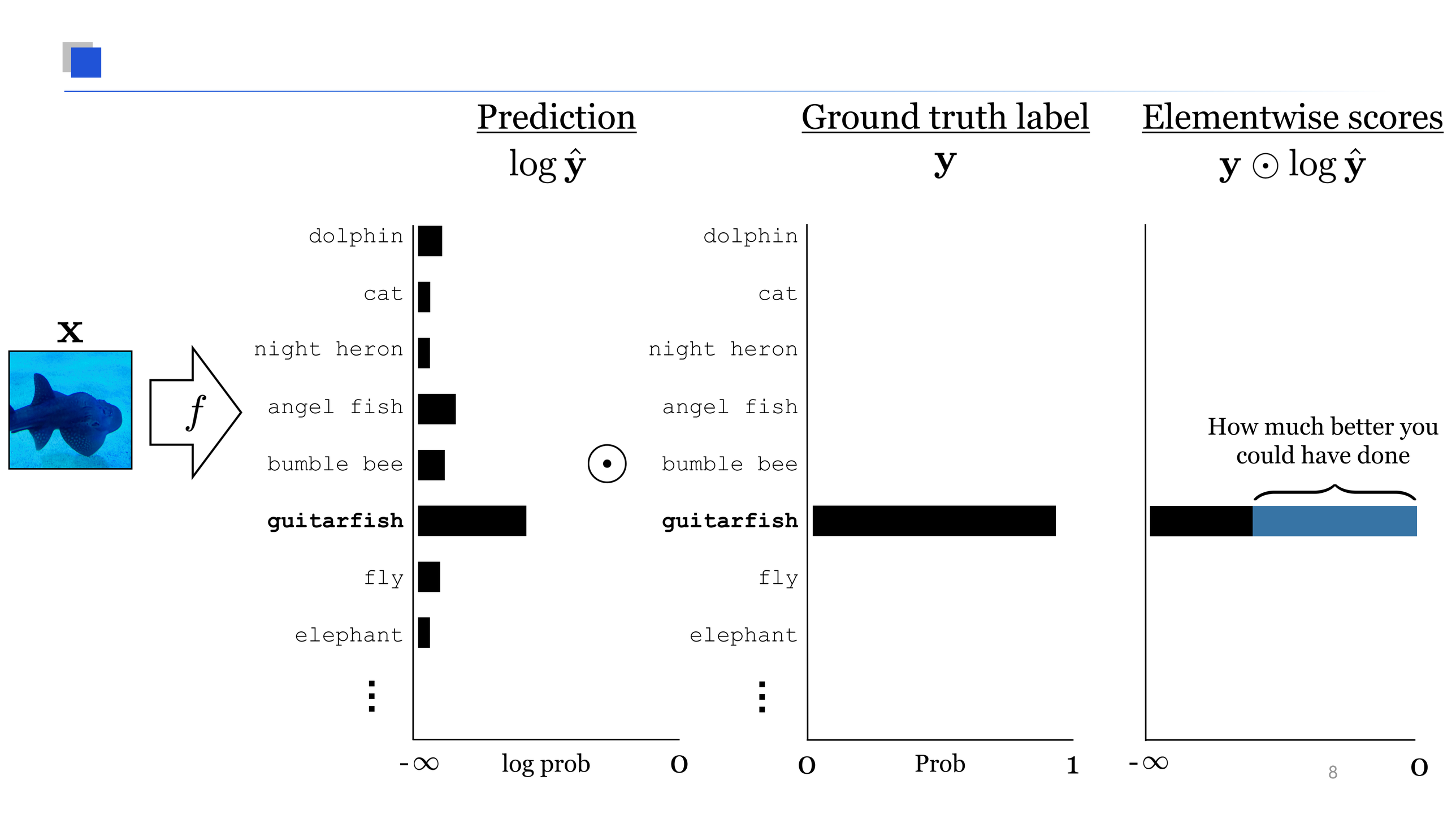


$$x_{out}[i] = \max(x_{in}[i], 0)$$



Fast Activations vs Slow Parameters




Prediction
log ŷ
Ground truth label
y
Elementwise scores
y ⊙ log ŷ
x
f
dolphin
cat
night heron
angel fish
bumble bee
guitarfish
fly
elephant
⊙
-∞
log prob
0
0
Prob
1
How much better you could have done
-∞
0

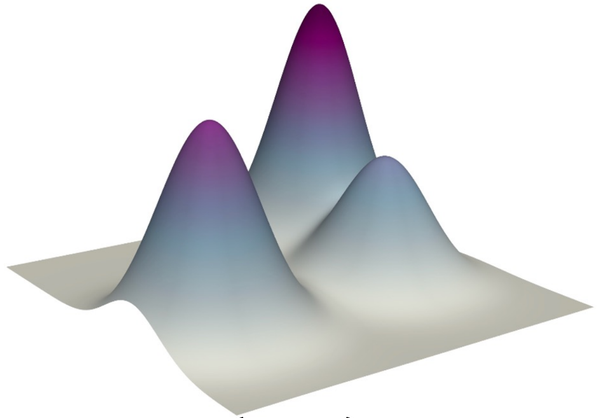
蒙特卡洛 (Monte Carlo)

估计期望： $\mathbb{E}_{X\sim p}[f(X)] = \int_{\mathbb{R}^d} f(\boldsymbol{x}) \cdot p(\boldsymbol{x})\, d\boldsymbol{x}.$

1. Draw N random samples from the probability distribution $p(x)$; denote them by $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_N$.
2. Calculate $\bar{\boldsymbol{f}} = \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{f}(\mathbf{x}_i)$.
3. Return $\bar{\boldsymbol{f}}$ as an approximation to $\mathbb{E}_{X\sim p}[f(X)]$.

$$\mathbb{E}_{X\sim p}[\boldsymbol{f}(\boldsymbol{X})] \approx \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{f}(\boldsymbol{x}_i)$$

经验风险近似期望风险

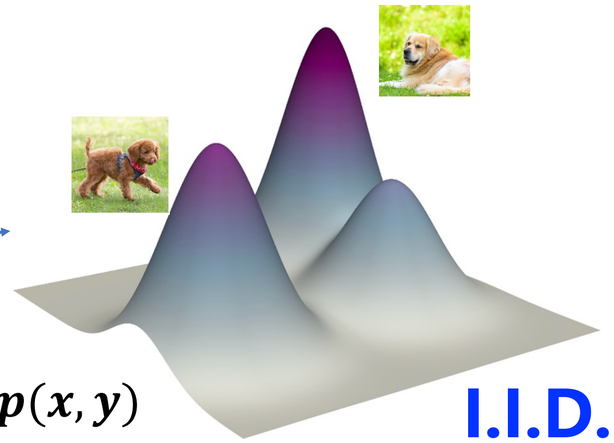


$p(x, y)$

真实的未知分布

采样

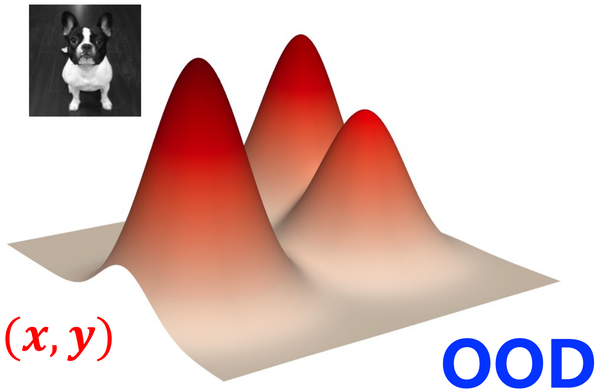
$\{x_n, y_n\}$



$p(x, y)$

I.I.D.

机器学习模型



$p'(x, y)$

OOD

$$\mathbb{E}_{p(x,y)}[\mathcal{L}(\hat{y}, y)]$$

期望风险

近似

$$\mathcal{L} = \frac{1}{N}\sum_{n=1}^{N}\mathcal{L}(\hat{y}, y)$$

经验风险

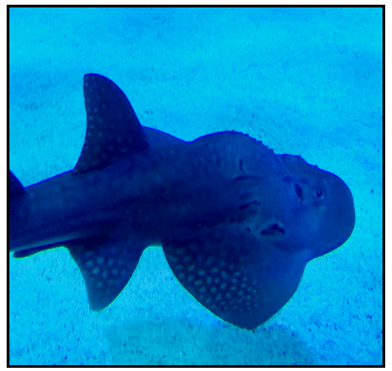
深度学习

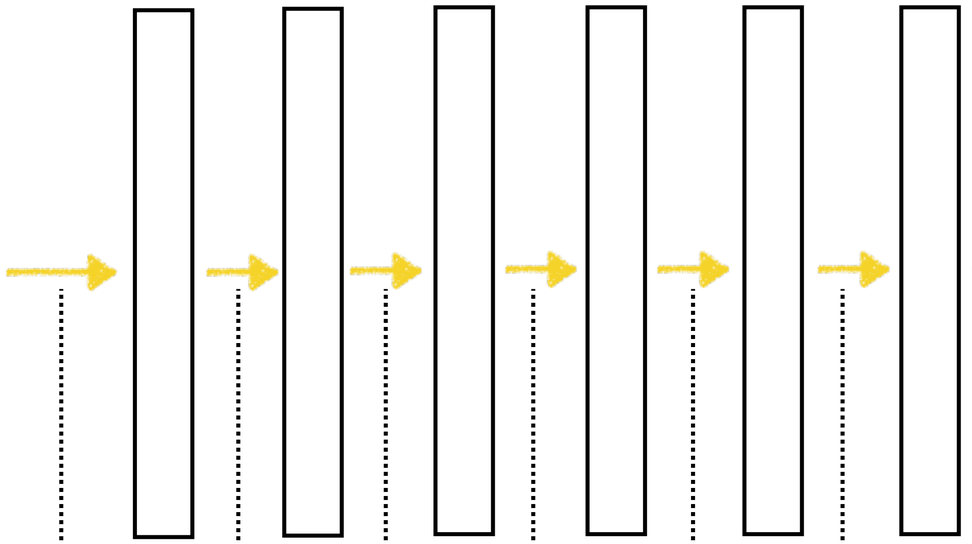
Learned

$\mathbf{y}^{(1)}$

guitarfish

$\mathbf{x}^{(1)}$





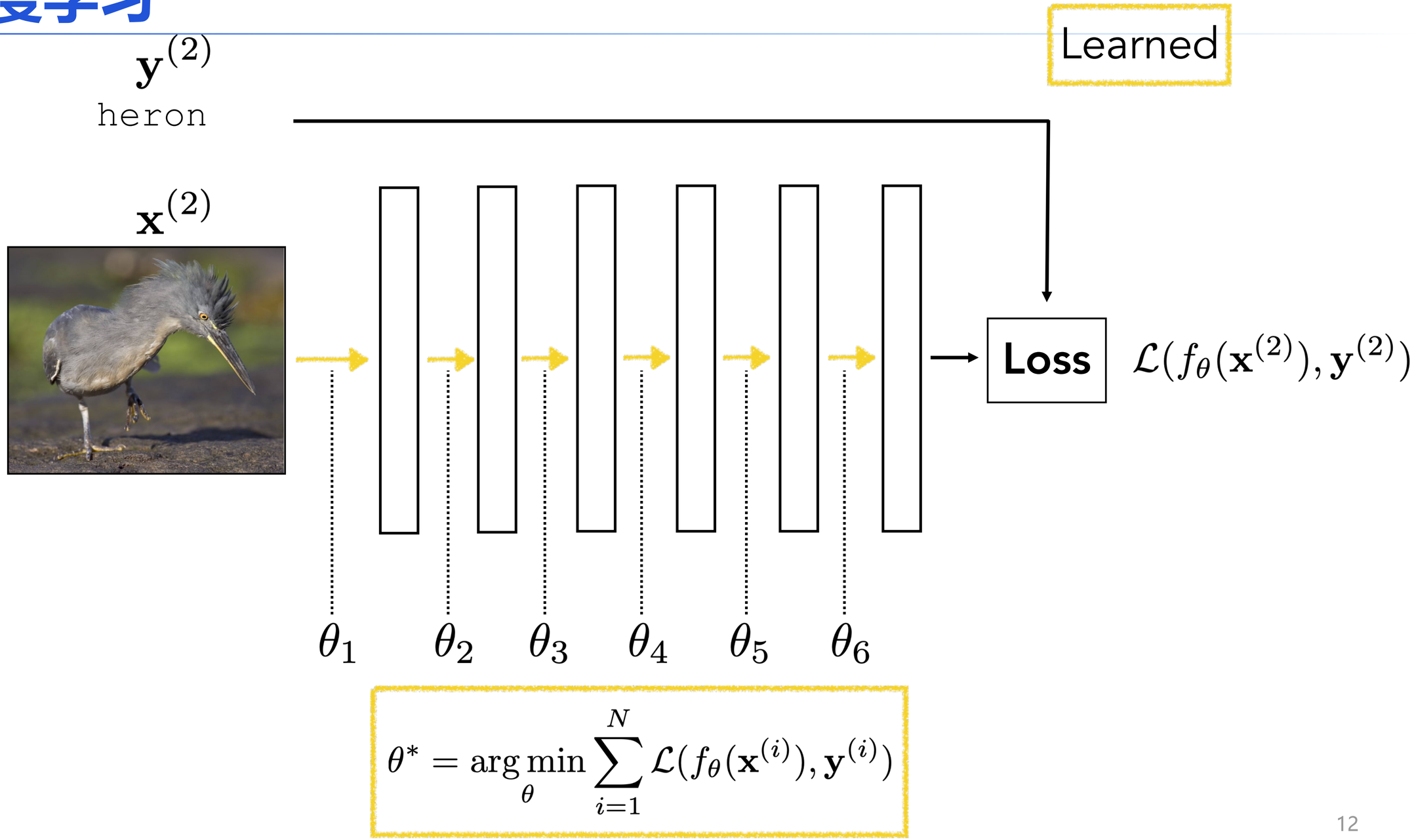
Loss

$\mathcal{L}(f_\theta(\mathbf{x}^{(1)}), \mathbf{y}^{(1)})$

θ_1 θ_2 θ_3 θ_4 θ_5 θ_6

$$\theta^* = \arg\min_\theta \sum_{i=1}^{N} \mathcal{L}(f_\theta(\mathbf{x}^{(i)}), \mathbf{y}^{(i)})$$

深度学习


$\mathbf{y}^{(2)}$
heron
Learned
$\mathbf{x}^{(2)}$
Loss
$\mathcal{L}(f_\theta(\mathbf{x}^{(2)}), \mathbf{y}^{(2)})$
θ_1
θ_2
θ_3
θ_4
θ_5
θ_6
$\theta^* = \arg\min_\theta \sum_{i=1}^{N} \mathcal{L}(f_\theta(\mathbf{x}^{(i)}), \mathbf{y}^{(i)})$

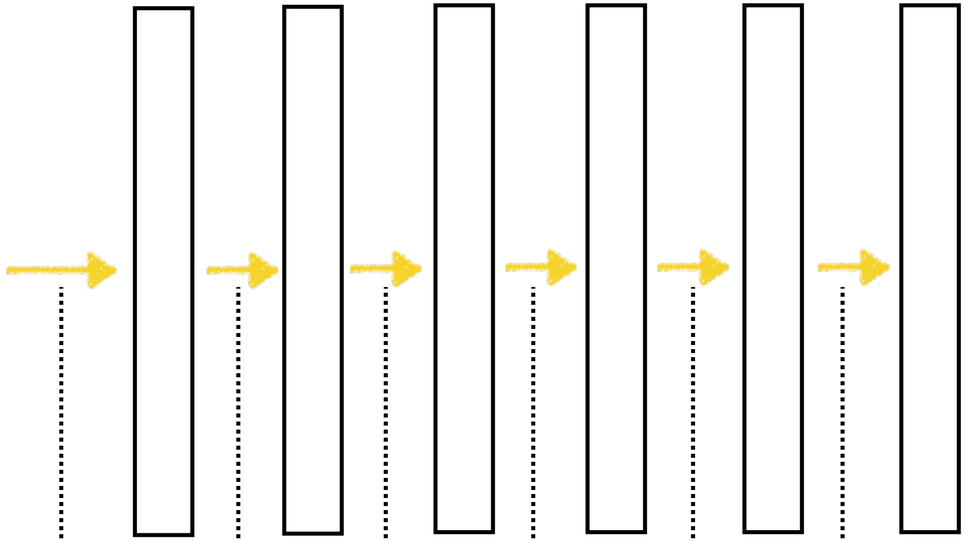
深度学习

Learned

$\mathbf{y}^{(i)}$

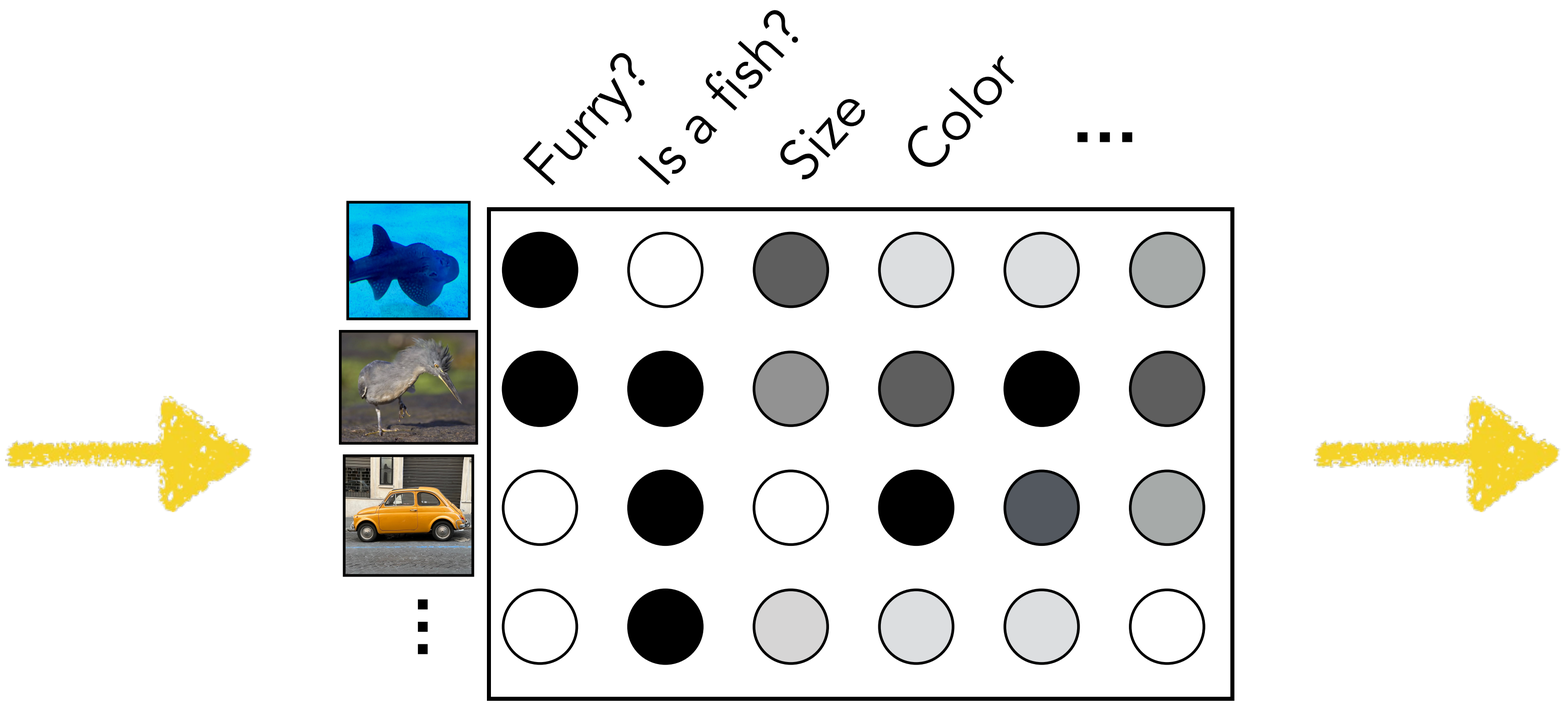
car

$\mathbf{x}^{(i)}$



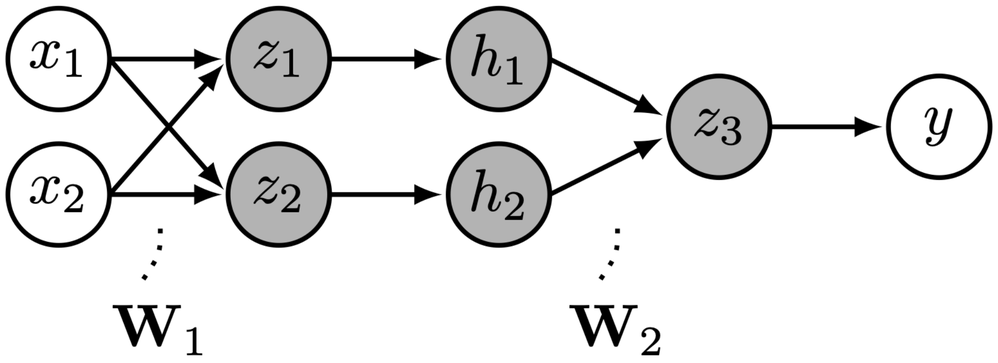
$$\theta^* = \arg\min_\theta \sum_{i=1}^{N} \mathcal{L}(f_\theta(\mathbf{x}^{(i)}), \mathbf{y}^{(i)})$$

张量表示



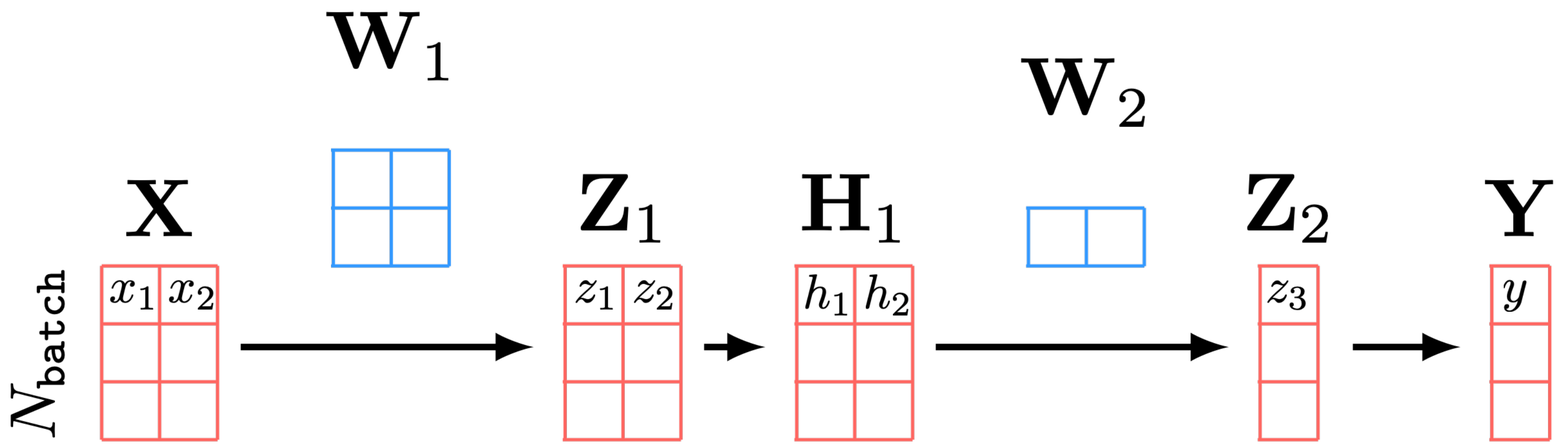
Each layer is a representation of the data

张量表示



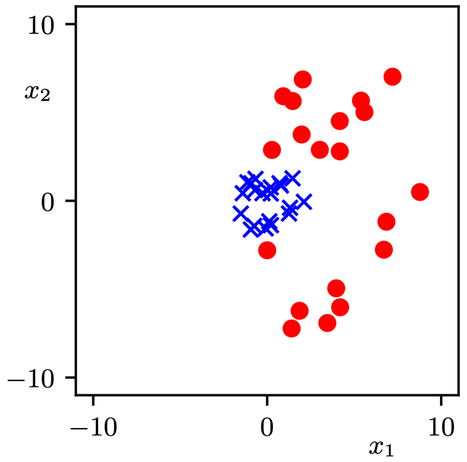
$$\mathbf{z} = \mathbf{W}_1\mathbf{x} + \mathbf{b}_1$$
$$\mathbf{h} = g(\mathbf{z})$$
$$z_3 = \mathbf{W}_2\mathbf{h} + b_2$$
$$y = 1(z_3 > 0)$$

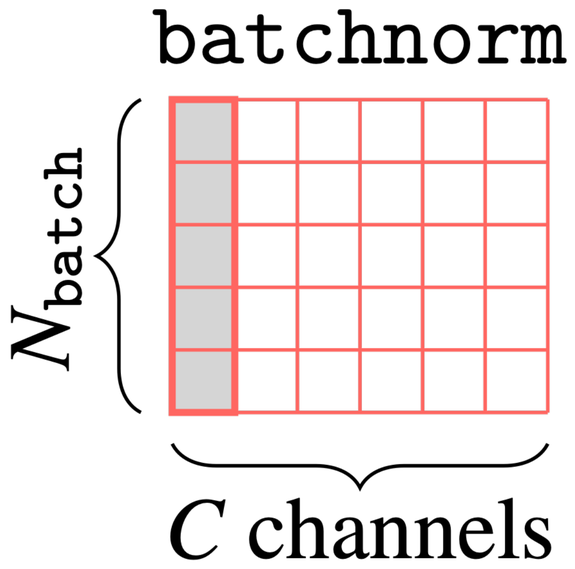
Tensor processing with batch size = 3:

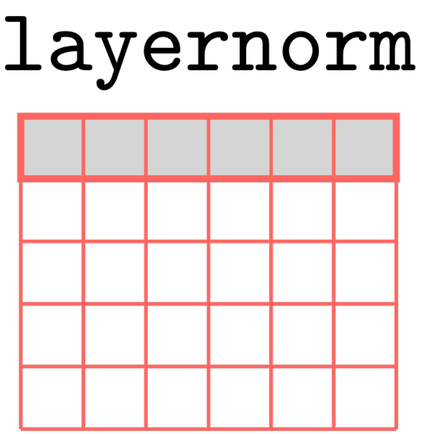


归一化层

- **归一化层基于神经元群体的值，将激活值压缩至更优的数值范围。**

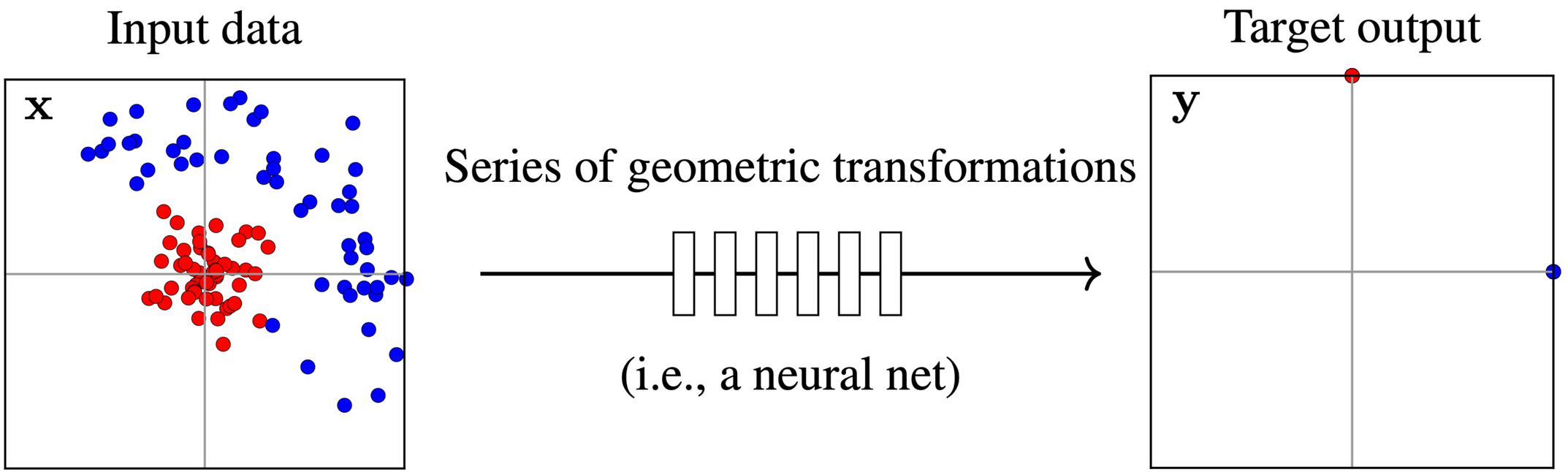




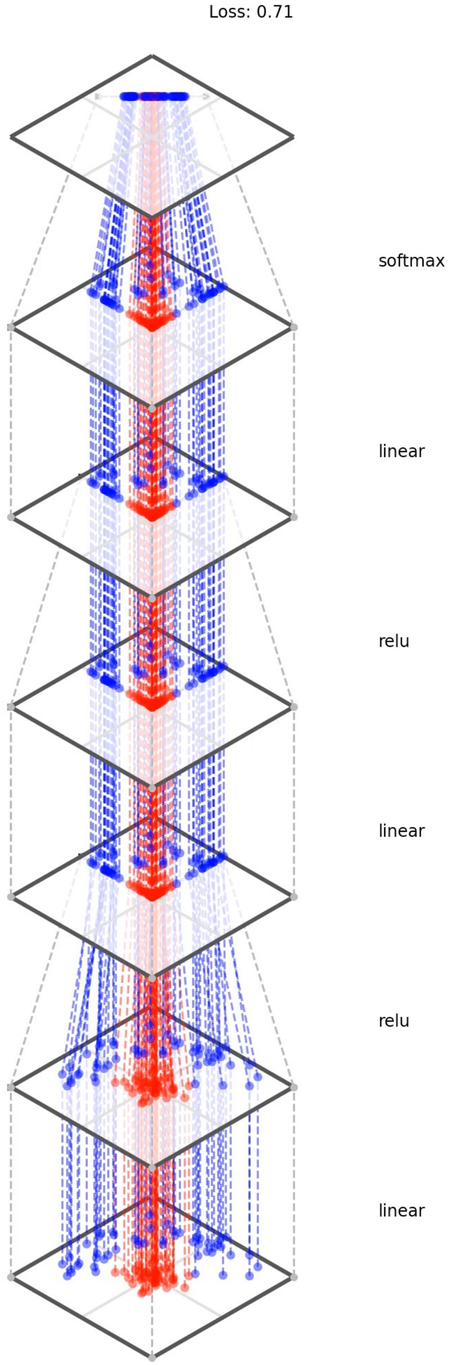


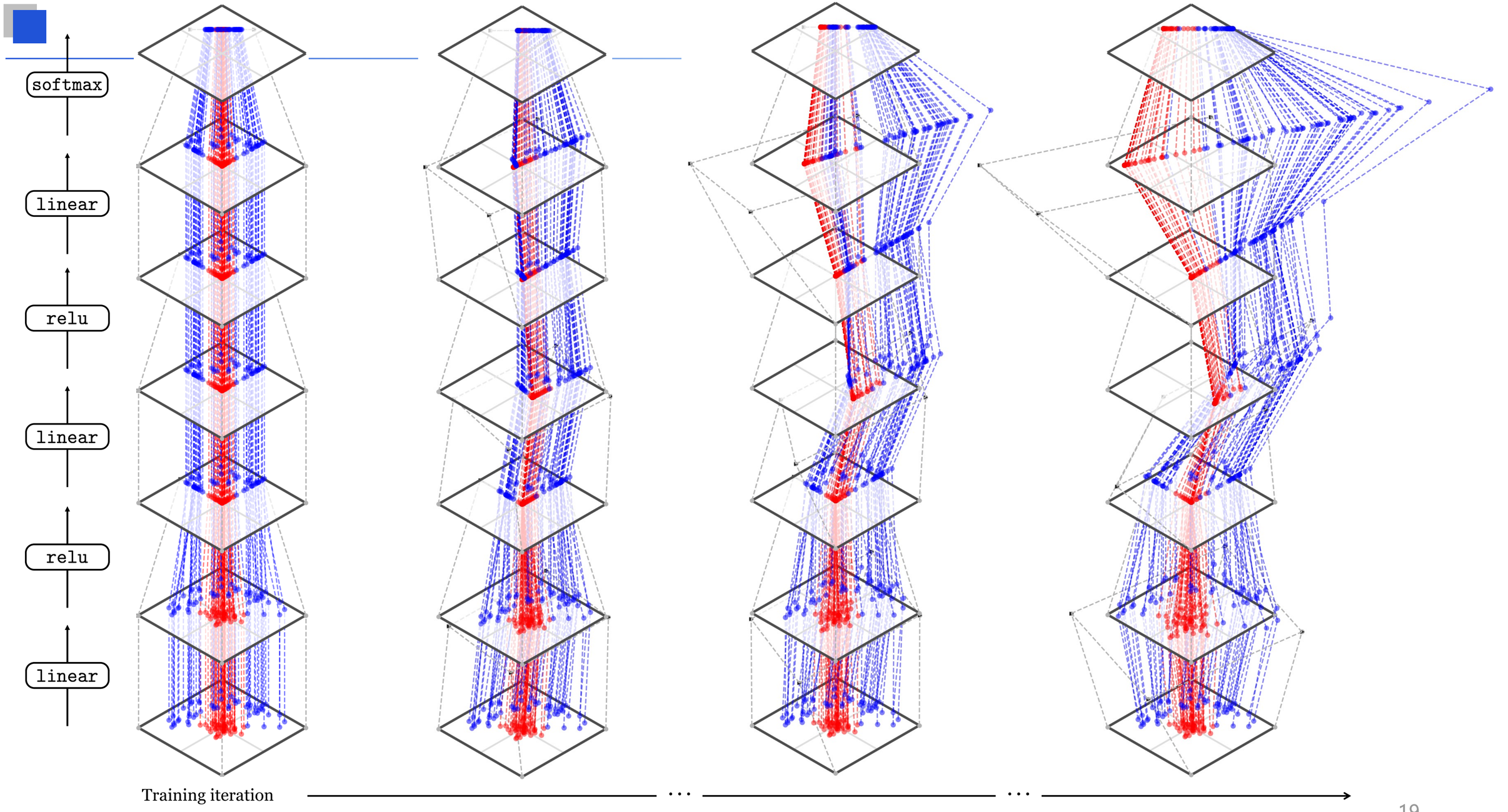
- **Data normalization:** each variable now has zero mean and unit variance across the data set.
- **Batchnorm:** standardizes all values in a column of the activation tensor (subtracts mean, divides by variance)
- **Layernorm:** does the same except to a row of the activation tensor.

What does training a deep net classifier look like?



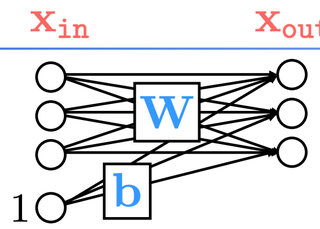
Steps of gradient descent training an MLP with hidden dimensionality 2 on each layer.



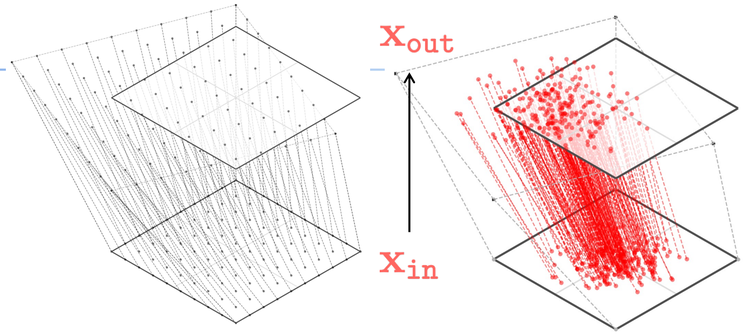

softmax
linear
relu
linear
relu
linear
Training iteration

Wiring graph | Equation | Mapping

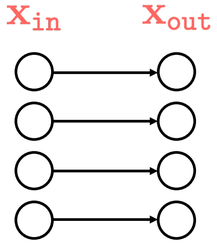
linear



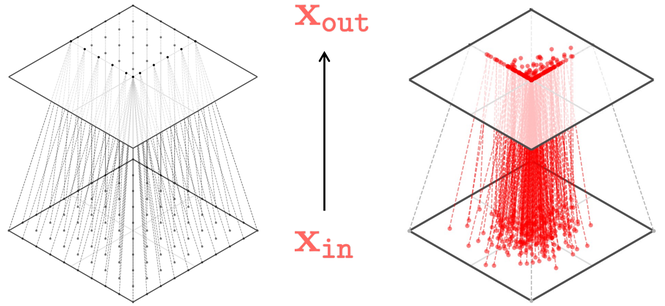
$$\mathbf{x}_{\text{out}} = \mathbf{W}\mathbf{x}_{\text{in}} + \mathbf{b}$$



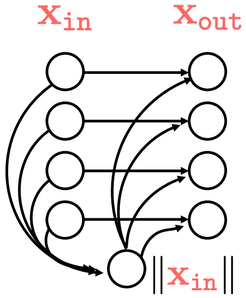
relu



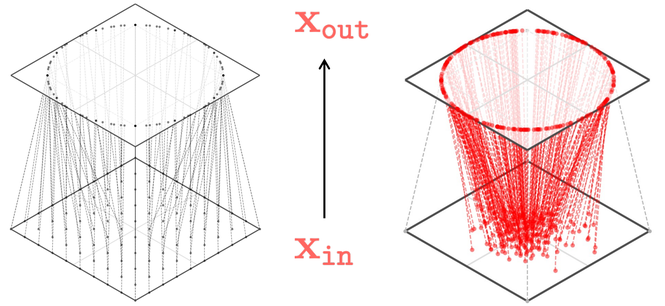
$$x_{\text{out}}[i] = \max(x_{\text{in}}[i], 0)$$



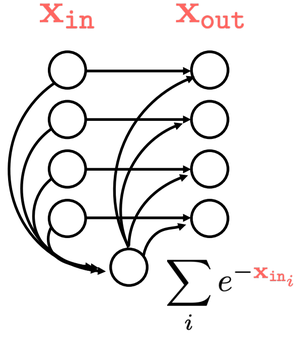
L2-norm



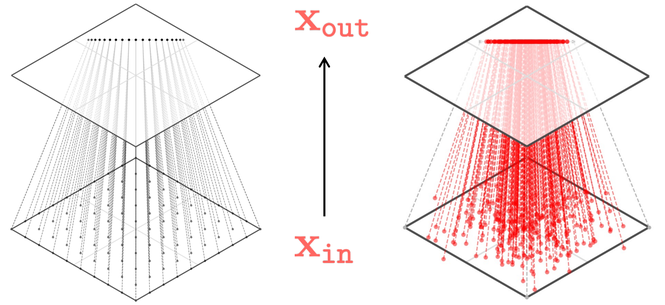
$$x_{\text{out}}[i] = \frac{x_{\text{in}}[i]}{\|\mathbf{x}_{\text{in}}\|_2}$$

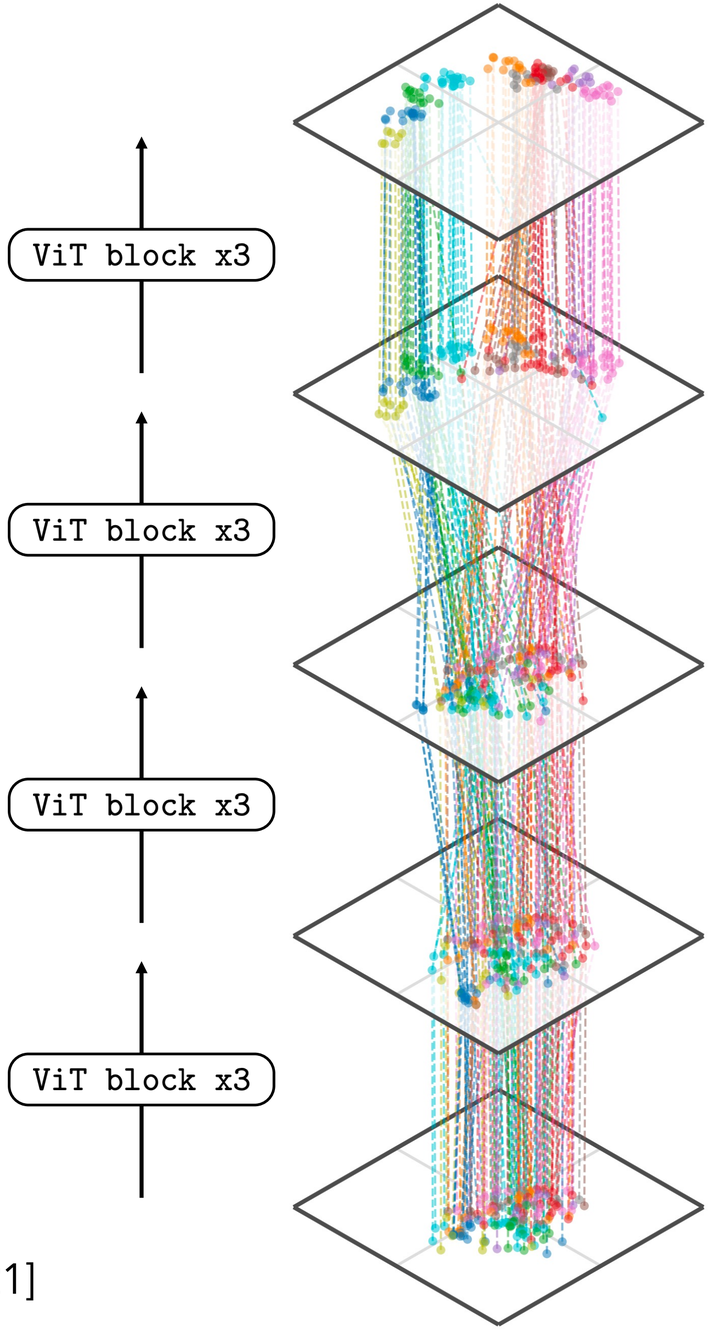


softmax



$$x_{\text{out}}[i] = \frac{e^{-\tau x_{\text{in}}[i]}}{\sum_{k=1}^{K} e^{-\tau x_{\text{in}}[k]}}$$

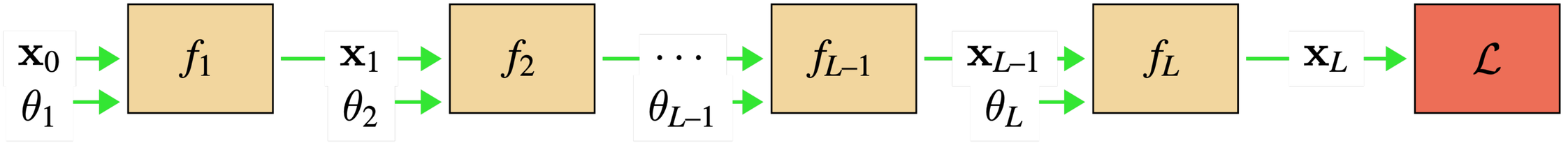




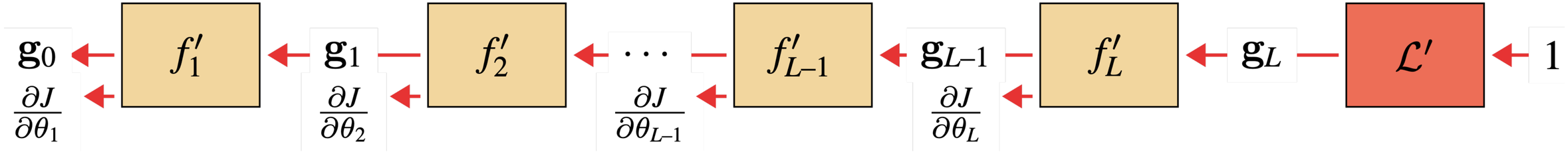
[**CLIP**, Radford*, Kim* et al., ICML 2021]

BP 算法

Forward:



Backward:



Update:

$$\theta^{i+1} \leftarrow \theta^i - \eta \left(\frac{\partial J}{\partial \theta} \right)^\mathsf{T}$$

… and repeat

Thank You